
Tonality and Transformation, by Steven Rings. *Oxford Studies in Music Theory*. Richard Cohn, series editor. New York: Oxford University Press, 2011. xiv + 243 pp. ISBN 978-0-19-538427-7. \$35 (Hardback).¹



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Steven Rings' *Tonality and Transformation* is only one entry in a deluge of recent volumes in the *Oxford Studies in Music Theory* series,² but its extension of the transformational ideas pioneered by scholars like David Lewin, Brian Hyer, and Richard Cohn into the domain of tonal music sets it apart as a significant milestone in contemporary music theory. Through the "technologies" of transformational theory, Rings articulates his nuanced perspective on tonal hearing by mixing in generous helpings of Husserlian philosophy and mathematical graph theory, and situating the result in a broad historical context. The text is also sprinkled throughout with illuminating methodological comments, which weave a parallel narrative of analytical pluralism by bringing transformational ideas into contact with other modes of analysis (primarily Schenkerian theory) and arguing that they can complement rather than compete with one another. Rings manages to achieve all this in 222 concise pages that will prove both accessible to students and satisfying for experts, making the book simultaneously a substantial theoretical contribution, a valuable exposition of existing transformational ideas, and an exhilarating case study for creative, interdisciplinary music theorizing.

[2] *Tonality and Transformation* is divided into two parts. The first three chapters constitute the theory proper – a reimagining of tonal music theory from the ground up, via the theoretical technologies mentioned above. The final 70 pages consist of four analytical essays, which begin to put Rings' ideas into practice. Two of the analyses (Brahms' Intermezzo Op. 118, No. 2 and String Quintet, Op. 111, ii) are carried over from Rings' eponymous 2006 Ph.D. thesis, while two (Mozart's "Un'aura amorosa" from *Così fan tutte* and the E-major Fugue from Book II of Bach's *Well-Tempered Clavier*) are completely new, replacing two Schubert analyses.³ Full, lightly-annotated scores of the music under consideration are helpfully included at the end of each chapter.

[3] While the book arises from a 2006 dissertation, it has been substantially revised and updated, accounting for new research by theorists such as Julian Hook, David Huron, and Dmitri Tymoczko. Perhaps the most notable addition to the original dissertation is the concept of a Tonal GIS (Generalized Interval System), which turns out to be one of the book's most significant theoretical set pieces. Intriguingly, this addition makes the book closely reflect the structure of David Lewin's *Generalized Musical Intervals and Transformations* (1987, hereafter *GMIT*) which, along with some earlier articles,

¹ I am grateful to Suzie Clark for her insightful reading of an early draft of this review, and to Duncan Schultz for his comments on this final version. Any errors in judgment or interpretation are my own.

² Other titles from the past year include Richard Cohn, *Audacious Euphony: Chromatic Harmony and the Triad's Second Nature* (2012); Janet Schmalfeldt, *In the Process of Becoming: Philosophical and Analytic Perspectives on Form in Early Nineteenth-Century Music* (2011); Gretchen Horlacher, *Building Blocks: Repetition and Continuity in the Music of Stravinsky* (2011); and Dmitri Tymoczko, *A Geometry of Music: Harmony and Counterpoint in the Extended Common Practice* (2011).

³ See Steven Rings, "Tonality and Transformation" (Ph.D. diss., Yale University, 2006). One of these Schubert chapters saw print as "Perspectives on Tonality and Transformation in Schubert's Impromptu in E-flat, D. 899," *Journal of Schenkerian Studies* 2 (2007).

inaugurated the field of transformational theory. Henry Klumpenhouwer has argued that the “long narrative” of *GMIT* is one that moves the reader from an intervallic orientation (reflected by the GIS methodology that dominates the first part of the book) to a more first-person position, characterized by the study of musical *actions* (reflected by transformational theory, which is introduced around the seventh chapter of *GMIT*).⁴ The formal layout of the first few chapters of *Tonality and Transformation* mirrors this narrative: Chapter One presents mathematical preliminaries, Chapter Two introduces a Tonal GIS, and Chapter Three explores the use of transformational networks in tonal analysis. But while Klumpenhouwer believes that Lewin’s text embeds a clear anti-Cartesian agenda, with transformation replacing GIS, Rings argues that as long as we are aware of the ideological baggage carried by any given methodology, we can feel comfortable about allowing both of these technologies and more to co-exist within our toolbox.⁵ Even so, the narrative parallels to Lewin are striking – it seems clear from the outset that *Tonality and Transformation* is a direct response to *GMIT*. It is as much a return to transformational “first-principles” as it is a re-interpretation and expansion of Lewin’s ideas for the exploration of new repertoire. On the very first page, Rings distances himself from the emphasis on voice leading efficiency that has come to dominate Neo-Riemannian and geometric music theories, and declares that he “seeks to return to certain fundamental ideas from transformational and GIS theory, exploring their potential to illuminate familiar aspects of tonal phenomenology.”⁶

The Tonal GIS

[4] *Tonality and Transformation* opens with an introduction to the mathematical principles that underlie transformational theory. Rings carefully walks the reader through the functions, sets, and groups that are relevant to diatonic and chromatic musical spaces, generally without *GMIT*’s sometimes forbidding technical language. He then explores the nature and applications of both GIS and transformational statements, outlining the strengths and weaknesses of each, and providing some simple examples. This aspect of the first chapter could serve (perhaps alongside Satyendra 2004) as introductory reading for graduate students or advanced undergraduates studying transformational theory for the first time. A detailed glossary, which goes into greater mathematical detail and even includes concepts that don’t make it into the text (such as Riemann’s *Schritt/Wechsel* system), a useful pedagogical addition. Along with mathematical preliminaries, the introductory chapter also places the book in a broad context of recent work in transformational theory (Hook 2007, Klumpenhouwer 2006, Tymoczko 2008 and 2009), and contains several significant disciplinary and meta-theoretical statements. I will address these broader concerns *after* my treatments of Chapters Two and Three.

⁴ See Henry Klumpenhouwer, “In Order to Stay Asleep as Observers: The Nature and Origins of Anti-Cartesianism in Lewin’s *Generalized Musical Intervals and Transformations*,” *Music Theory Spectrum* 28, no. 2 (2006): 277–281.

⁵ While methodological pluralism is an *idée fixe* of the book, Rings specifically addresses Klumpenhouwer’s interpretation of Lewin in note 22 on page 17 of *Tonality and Transformation*.

⁶ See Rings, *Tonality and Transformation*, 1. While Neo-Riemannian theory and transformational theory are often grouped together in discussions (perhaps due to their common origins), Rings is correct in insisting throughout the book that they have different goals, and should be treated independently. It is also worth noting that two other recent entries in the Oxford Studies in music theory (Richard Cohn’s *Audacious Euphony* and Dmitri Tymoczko’s *A Geometry of Music*) both distance themselves from the “Neo-Riemannian” label as well.

	$\hat{1}$	$\hat{2}$	$\hat{3}$	$\hat{4}$	$\hat{5}$	$\hat{6}$	$\hat{7}$
e							
t							
9							
8							
7							
6							
5							
4							
3							
2							
1							

Figure 1: The space S of 84 ordered pairs from Rings' Tonal GIS⁸

[5] Chapter Two introduces one of the book's theoretical centerpieces, the Tonal GIS. This GIS treats the notes heard in a tonal piece as ordered pairs, in the form (sd, pc). The first number indicates a scale degree, which Rings christens a *qualia*, in a locution borrowed from the philosophy of mind. Such an attribution is not a feature of the music itself, but is supplied by the listener, who mentally situates the pitch at hand within some scale.⁷ The second member of the ordered pair represents the acoustic signal of the note, most often written in *Tonality and Transformation* with familiar pitch-class notation, as an integer mod 12. The ordered pair ($\hat{1}$, 5), for example, represents the pitch-class F heard as the tonic, while ($\hat{4}$, 5) represents the same pitch-class, this time heard as the fourth scale degree in the key of C (either major or minor). The full set of 84 possible ordered pairs constitutes the space S of the tonal GIS. The *group* of 84 intervals is represented in much the same way, as an ordered pair (sdint, pcint). The first term, sdint, is a familiar tonal interval such as a 2nd or a 3rd, while the second term, pcint, is a simple directed pitch-class interval, representing the number of half-steps traversed. Formally, these 84 ordered pairs constitute the group IVLS for the tonal GIS. These two formal elements yield the space depicted in Figure 1, in which any two (or more) tonal pitch coordinates can be related by any of the 84 intervals. Similar diagrams reoccur throughout the book in various forms – with flipped axes, with pitch-classes in reverse order, or occasionally truncated so as to show only the relevant portion of the space. At various points, the grid space is used to illustrate static intervals, the movement of melodies, and the possible voice-leading between chords.

[6] After the reader has had a chance to become comfortable with the ordered pair notation, Rings begins to pile on more functions, and it is here that the undergrowth of formalism grows a little dense. First, Rings defines *key* in the GIS as an ordered septuple, so that C major is written as (0, 2, 4, 5, 7, 9, e). Next,

⁷ Rings' principal influence here is David Huron, who argues in Chapter 9 of *Sweet Anticipation: Music and the Psychology of Expectation* (Cambridge, Mass.: MIT Press, 2006) that listener perception of scale-degree qualia arises from statistical learning rather than theoretical training.

⁸ This grid is modeled after the one on page 45 of *Tonality and Transformation*. Since both parameters of the grid are conceptual loops (which structure the grid as a torus, like the famous Riemannian *tonnetz*) the grid can also start on a different pc in order to depict a different area of pitch space, as shown on page 69. Since a blank grid never appears in the book, I have modified this one in order to depict the empty space, so that readers may examine it on their own.

the function acc_n attempts to account for the phenomenon of hearing accidentals as modifications of a given diatonic step n . Thus, for example, a pc 8 could be heard in the context of the C major ordered septuple as a raised $\hat{5}$. Acc_5 would then process this tone in relation to pc 7 (the diatonic pitch-class associated with $\hat{5}$) by measuring the interval between the diatonic pitch and its modification, through the function $\text{acc}_5(8) = \text{int}(\hat{7}, 8) = 1$. The resulting integer 1 indicates that the pitch has been raised by one semitone. This result can be translated (by a matrix like the one shown in Rings' Figure 2.27, on page 73) into a simple # sign, yielding the ordered pair ($\#5$, 8).

[7] This process is, on some level, necessary – a *tonal* GIS must be able to account for keys in order to model tonal centrality. The method, however, is extremely cumbersome. It comes across as a complicated apparatus that exists solely to close a formal loophole in the theory, and contributes little to the musical experiences that Rings is otherwise concerned with. A mathematical function that laboriously carries out an act of tonal hearing, in search of a single answer, feels out of place in a book that explicitly concerns itself with the multivalent nature of listening. Additionally, as a purely formal component, acc_n seems flawed in its conception, for it relies upon the theorist/listener to supply the initial scale degree interpretation – the apperception that a given chromatic pitch is to be heard as an altered scale degree n . On paper, the example detailed in the previous paragraph could just as easily examine pc 8 as $\flat 6$, formalized as $\text{acc}_6 = \text{int}(\hat{9}, 8) = 11$ (or, taking advantage of mod-12 pc space, -1). However, acc_n makes no judgment about which is correct, and offers the theorist no insight on which to choose. The function surely imposes certain limits – since, following Occam's Razor, a "hearing" that produces a large interval involving multiple flats will almost certainly be discarded in favor of a simpler interpretation – but it cannot judge on its own whether $\#5$ or $\flat 6$ is to be preferred. This is not a huge failing, since Rings is clear throughout the book that his formalisms are designed to augment the apperceptions of the listener, never to replace them. But perhaps it is telling that, after their initial exposition, key septuples and the acc_n function appear only very rarely throughout the rest of the book, and never again at center stage.

[8] With his GIS technology in place, Rings rethinks many of the pillars of tonal theory. Shifts in tonal centrality, for example, with their accompanying reshuffling of qualia, are represented by *pivot intervals* – intervals between ordered sd/pc pairs that take the form $(n, 0)$, where n represents some distance in scale-degree (quale) space, and 0 indicates that the pitch-class stays the same. Modulation between keys is conceived as a transformation between ordered septuplets (the diatonic sets described above), achieved by multiple simultaneous pivots. Figure 2 depicts a modulation from C to G, which consists of six simple pivots, through which every note except F moves by $(4\text{th}, 0)$, and one *skew pivot*, in which a single member of the diatonic collection (F) moves by half-step to become a member of the new diatonic collection ($F\sharp$). The skew pivot in this particular figure, $(4\text{th}, 1)$, embodies the contradiction inherent in a modulation from I to V. While only a single note (F) changes, by only a single half step (to $F\sharp$), this incremental movement initiates a seismic shift, moving each quale by a fourth – as far as is possible, under the mod-7 arithmetic of diatonic space, before coming back around. The operation is simultaneously elegant and clumsy, perhaps by design. The bent arrow shows quite intuitively that only a tiny change in pitch content takes place: $\hat{4}$ is subtly "bent" up by a half step. The barrage of additional arrows, however, underscores just how dramatically the tonal qualia are shuffled in even the simplest modulation, from the dark gray of C major to the light gray of G major. The same pitch class, in a

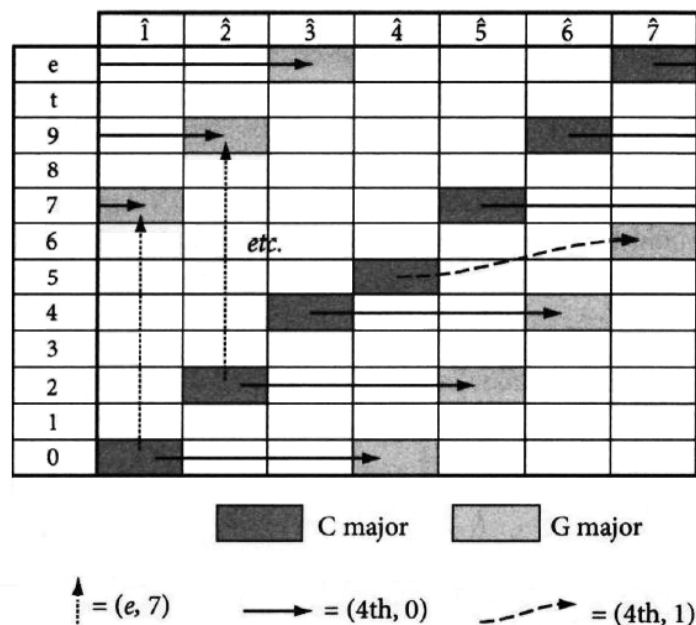


Figure 2: Transformation from C Major to G Major (Rings 2011, 67). The “bent” arrow indicates the skew pivot.

shifting tonal context, can embody two completely different phenomenological qualities in the space of only a few seconds. The stable tonic of C major ($\hat{1}, 0$), for example, could reappear a bar or two later as a tonally-charged ($\hat{4}, 0$) in need of resolution – the same acoustic signal, but emphatically not where we last mentally “left it.” This complete reorientation captures the sense that a pitch class associated with one quale can take on an entirely different character very quickly; the grid notation shows how the shifting qualia of modulation make the same acoustic signal sound quite “far” from its previous iteration.

Tonal Transformation and Intentionality

[9] Although Rings wears the methodological and orthographic influence of Lewin’s transformational theory on his sleeve, it becomes apparent in the course of reading *Tonality and Transformation* that the third word of the title could just as easily have been *intentionality*, an idea that is central to Rings’ philosophy of tonality. He conceives of intentionality in a roughly Husserlian manner: it is the act of directing one’s consciousness toward an object. For Husserl, when we attend to an object mentally, we imbue it with *intentional content* (or *noema*): our own interpretation of its nature or significance. A glass of water is not intended simply as a glass; it is “half-empty,” or “half-full,” or perhaps “within arm’s reach” or “too warm to drink.”⁹ In much the same way, Rings argues that when we hear tonally, we imbue acoustic objects with intentional content by relating them to a tonic pitch or triad. For Rings, this

⁹ A concise introduction to this aspect of Husserl’s thought can be found in David Woodruff Smith, *Husserl* (New York: Taylor and Francis, 2007), 56–59 and 206–210.

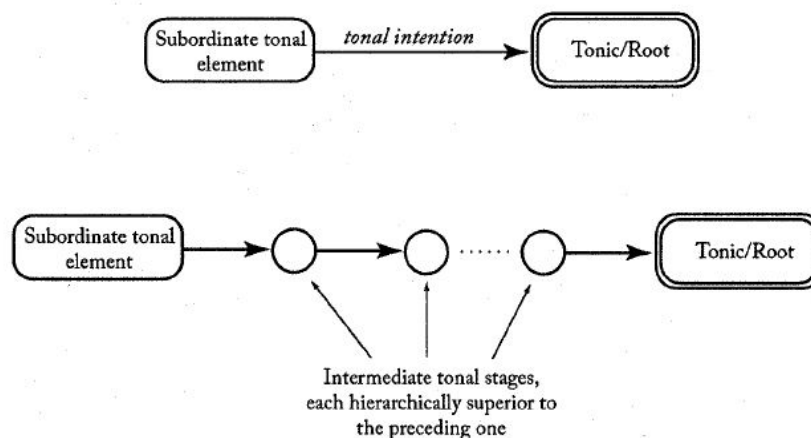


Figure 3: *Some Generic Oriented Digraphs* (Rings 2011, 106)

intentional content takes the form of a particular tonal quale, which positions each tone in relation to the rest of its diatonic context.¹⁰

[10] Chapter Three adds a visual interpretation of tonal intentionality, by way of the book's other theoretical pillar, the *oriented digraph* (directed graph). Oriented digraphs are "configuration[s] of nodes and arrows, all of whose arrows are directed toward a single node," as schematized in Figure 3.¹¹ This single node is known as the *root node*—it represents the tonic (on either a local or global level), and is set off in Rings' notation by a double border. This double border, visually distinguishes Rings' tonal digraphs from any number of non-tonal digraphs in other transformational studies, and provides the tonal center of each diagram. Numerous other requirements govern oriented digraphs: all nodes must be connected, all source and intermediate nodes must eventually lead to the root node, and the path through the nodes must be unidirectional, with no loops or two-way arrows.

[11] Rings uses digraphs to explore the intentional nature of tonal hearing. The arrows in Fig. 3 represent Rings' notion of tonal intentionality; he writes: "we can understand [the] arrows to model one of the things one 'does' to a sounding entity in the process of hearing it as tonal."¹² Thus, the right facing arrow in Fig. 3a indicates that the *source node*, the "subordinate tonal element," is heard/intended through its relationship to the tonic/root node. Fig. 3b represents a similar relationship, mediated by several additional nodes between the network's source node and its goal. Movement through these diagrams (which can be arranged as either *event networks*, representing linear movement through a musical passage, or *spatial networks*, representing an atemporal arrangement of various tonal entities) is represented by the idea of tonal-intentional *energy*, which functions as a representation of stability in the networks of Chapter Three. An arrow departing from a given node invests that node with some degree of

¹⁰ David Huron has taken the concept of tonal qualia even further by exploring the informal descriptions of stability and instability that tend to accrue around certain scale degrees. See *Sweet Anticipation*, 144–147, especially the table on page 145.

¹¹ Rings, *Tonality and Transformation*, 110.

¹² Rings, *Tonality and Transformation*, 105.

instability; intentional energy flows *away* from it. The root node, as the only node in the digraph that has no arrows pointing away from it, is the most stable.¹³

[12] Rings' notion of energy places him in a productive dialogue with several music-theoretical traditions, including Rameau's metaphors of "tonal gravity," Kurth's notion of "tonal energy," and Schenkerian tales of organic tones that grow and strive. Some of the most compelling pages of Chapter Three are Rings' parallel readings of several of these theorists, through which he demonstrates both his relationship to earlier tonal theories, and the ability of tonal-intentional networks to subsume, clarify, and even amplify their ideas.¹⁴ Furthermore, Rings jettisons the less-fashionable metaphysical commitments of these theorists, reframing them in terms of intentionality and thus displacing the agency that strives for tonal resolution from the music to the listener. *Tonality and Transformation* is focused throughout on the *esthetic* aspects of music—those that involve the experience of the listener—as opposed to the *immanent* aspects, the structural properties of the music as written.¹⁵

[13] The rest of Chapter Three demonstrates how oriented digraphs can be used to model any number of tonal apperceptions, from the movements of individual pitches or voices, to entire networks of chords and transformations. Perhaps their most valuable contribution is their ability to model the *kinetic* nature of tonal hearing. Rings' root nodes (as he himself points out on page 115), feel anchored on the page, and often provide a fulcrum around which one can orient the subtle physicality of tonal listening in a way that is often absent from complex, non-oriented transformational graphs and networks.

Tonality and Transformation as Disciplinary Statement

[14] Having explored the two significant theoretical chapters, we will now examine how Rings frames his work. Interspersed among the preliminary remarks of the first chapter, which are designed to gently introduce the student or non-specialist to transformational theory, are several significant meta-theoretical statements that will be of more interest to professional theorists. Taken together, these statements not only situate *Tonality and Transformation* within the continuing reception of David Lewin's ideas, but contribute to larger and longer-running disciplinary debates about the relationship between analytical systems and repertoires. A brief, page-long aside on Cartesian dualism is a welcome recapitulation of Henry Klumpenhouwer's "In Order to Stay Asleep as Observers" (one of the more significant pieces of recent, philosophically-oriented Lewin reception), and hints at the larger issues at stake in one's choice of analytical "technology." Perhaps even more useful is Rings' treatment of Lewin's notion of *intuition*. Through a brief survey of textual examples, he argues that, for Lewin, intuitions are (a) culturally conditioned, and (b) able to be "sharpened, extended, or altered through analytical reflection."¹⁶ This leads Rings to propose that what Lewin calls intuition might be better described,

¹³ Ideas of stability and instability are also treated in the discussions of "resolving transformations." See Rings, *Tonality and Transformation*, 125–128.

¹⁴ See, for example, pages 107–110 (on Rameau, Riemann, Zuckerkandl, and Schachter) and 116–121 (on Riemannian functions), and 144–148 (on Schenker).

¹⁵ Rings first refers to these concepts on page 36, having drawn them from the first chapter of Jean-Jacques Nattiez, *Music as Discourse: Toward a Semiology of Music*, trans. Carolyn Abbate (Princeton: Princeton University Press, 1990).

¹⁶ Rings, *Tonality and Transformation*, 17.

following William James, as *apperception*: a process of perception that is “influenced by past experience and may involve present reflection.”¹⁷ Much unexplored territory remains for further discussion, and while various footnotes hint that Rings has the philosophical chops (and perhaps some unspoken opinions) with which to carry the argument forward, he chooses to leave the issue there; further exploration of the problematic concept of “musical intuitions” will have to wait for another day.

[15] Finally, Chapter One includes a brief excursus into Schenkerian theory. Schenkerian analysis and Neo-Riemannian/transformational theories are often positioned as antagonists—separate theories that address completely discontinuous repertoires, and which have neither the ability nor the inclination to learn from one another. Rings writes passionately and persuasively against this state of affairs, arguing that “any tension or competition between the two methodologies is misplaced and unnecessary...they are *not* competing forms of the same kind of music theory, but represent distinctly different styles of music-analytical thought.”¹⁸ He demonstrates this by comparing two possible analyses of the first several measures of the famous Prelude to Bach’s Suite for Solo Cello, No. 1, in G major, shown in Figures 4 and 5. Rings argues that the Schenkerian analysis (Fig. 4) is a way of making certain statements about the *structure* of the music: it is underlaid by three-voice counterpoint, it composes out the tonic *Stufe*; the *Kopfton* is B, and so forth. The collection of transformational diagrams (Fig. 5), however, depicts multiple ways of hearing or conceptualizing the opening leap: it can be heard as moving between two members of the same scale (Fig. 5b, in which the interval $i = +9$), two members of the same major triad (Fig. 5c, in which $i = +4$), two frequencies on the overtone series (Fig. 5d, in which $i = 5:2$), or any number of other interpretations. These interpretations, as discussed above, are *esthetic* rather than immanent—they pertain to our listening experience rather than the structure of the musical work.

[16] The differences between the two diagrams illustrate one of the guiding principles of *Tonality and Transformation*: different theories can be used, at will and in a non-competitive manner, to make different kinds of statements about music.¹⁹ At the conclusion of the discussion, Rings ultimately positions his book as a “prismatic” analytical approach, in which “phenomenologically rich local passages are refracted and explored from multiple perspectives.”²⁰

¹⁷ Rings, *Tonality and Transformation*, 18. An extended passage from James, partially quoted by Rings in his dissertation, explores the notion of apperception more fully, and is suggestive for the musical matter at hand. “Every impression that comes in from without, be it a sentence which we hear, an object of vision, or an effluvium which assails our nose, no sooner enters our consciousness than it is drafted off in some determinate direction or other, making connection with the other materials there ... The particular connections it strikes are determined by our past experiences and the ‘associations’ of the present sort of impression with them ... The impression arouses its old associates; they go out to meet it; it is received by them ... It is the fate of every impression thus to fall into a mind preoccupied with memories, ideas, and interests, and by these it is taken in. Educated as we already are, we never get an experience that remains for us completely nondescript; it always *reminds* us of something similar in quality, or of some context that might have surrounded it before, and which it now in some way suggests.” See William James, *Talks to Teachers on Psychology* (Cambridge, Mass.: Harvard University Press, 1983), 95–96.

¹⁸ Rings, *Tonality and Transformation*, 35.

¹⁹ In fact, Schenkerian statements can even be formulated in Rings’ transformational orthography, as shown on pages 147 and 148 of *Tonality and Transformation*.

²⁰ Rings, *Tonality and Transformation*, 38.

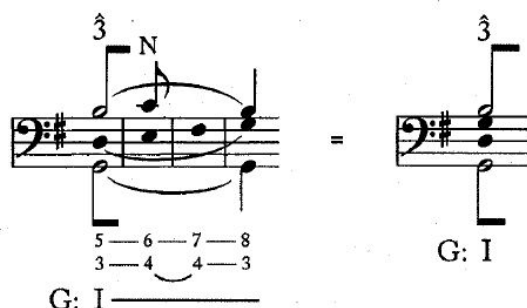


Figure 4: Schenkerian perspectives on Bach, Suite for Solo Cello No.1 in G major (Prelude), measures 1-4 (Rings 2011, 36)



Figure 5: Transformational perspectives on Bach, Suite for Solo Cello No. 1 in G major (Prelude), measure 1 (Rings 2011, 22)

[17] The final four chapters of the book execute this disciplinary agenda with panache, freely mixing methods when necessary in the search for musical insight. In both the theoretical and analytical chapters, Rings shows himself to be quite the polymath, as at home interpreting the metrical details of Italian poetry (see Chapter 5) as he is detailing the finer mathematical points of GIS theory. He also demonstrates great facility in jumping between contemporary and historical ideas, and in juxtaposing technical language with more subjective interpretation. For example, as mentioned earlier, several extended passages in Chapter Three explore variations on the idea of tonal intention in the writings of Jean-Phillipe Rameau, Hugo Riemann, Victor Zuckerkandl, and others. Chapter Six explores how the distinctions between thoroughbass and fundamental bass theories can affect our hearing, and constructs many of its transformational networks from ratios between harmonies, an ancient technique that Rings comes to by way of Lewin.²¹ Chapter Four (Bach's E-major Fugue) falls a bit flat, presenting a blow-by-blow account of the piece that lacks a punchy synthesis, but it still manages to resonate meaningfully with both Johann Joseph Fux's *Gradus ad Parnassum* and with a Greek-influenced tetrachordal theory, and to use both spatial and event networks to reveal interesting connections between the fugue's subject and countersubject. Chapter Five ("Un'aura amorosa," from Mozart's *Così fan tutte*) is perhaps the book's most wide-ranging and satisfying analysis, collecting ideas from historical theorists, (Kirnberger),

²¹ See David Lewin, *Generalized Musical Intervals and Transformations*, revised ed. (New York: Oxford University Press, 2007), 169–174.

Figure 6 consists of eight sub-diagrams (a-h) illustrating different analytical approaches to the opening motto of Brahms' String Quintet in G major, op. 111, ii, measure 1.

- (a) **D-minor hearing**: Musical notation in D minor (two flats). Below the staff, the Roman numeral analysis is $Dm: \hat{3} \text{ — } \hat{2} \text{ — } i \text{ — } V\sharp \text{ —}$.
- (b) **A-major hearing**: Musical notation in A major (three sharps). Below the staff, the Roman numeral analysis is $A: \flat 6 \text{ — } \hat{5} \text{ — } iv\flat \text{ — } I \text{ —}$.
- (c): Musical notation with fingerings (3, 2, 2, 1) indicated above the notes. Below the staff, the Roman numeral analysis is $Dm: i \text{ — } V\sharp$.
- (d): Musical notation with $\flat 6 = N$ and $\hat{5}$ indicated above the notes. Below the staff, the Roman numeral analysis is $A: iv\flat \text{ — } I$.
- (e): Schenkerian notation showing a progression from $^{\circ}T$ to $D^{\bullet}$. The diagram shows $D-$ in a circle pointing to $D^{\bullet}$, which is in a circle pointing to $A+$ in a circle.
- (f): Schenkerian notation showing a progression from $^{\circ}S$ to $T^{\bullet}$. The diagram shows $D-$ in a circle pointing to $S^{\bullet}$, which is in a circle pointing to $A+$ in a circle.
- (g): Tonal GIS sets for D minor. The diagram shows a progression from $\left\{ \begin{matrix} (\hat{3}, F) \\ (\hat{1}, D) \\ (\hat{5}, A) \end{matrix} \right\}$ to $\left\{ \begin{matrix} (\hat{2}, E) \\ (\flat 7, C\sharp) \\ (\hat{5}, A) \end{matrix} \right\}$ via the operation $(2nd^{-1}, -1)$. Below the sets are the Roman numeral analyses $d: i$ and $d: V$.
- (h): Tonal GIS sets for A major. The diagram shows a progression from $\left\{ \begin{matrix} (\flat 6, F) \\ (\hat{4}, D) \\ (\hat{1}, A) \end{matrix} \right\}$ to $\left\{ \begin{matrix} (\hat{5}, E) \\ (\hat{3}, C\sharp) \\ (\hat{1}, A) \end{matrix} \right\}$ via the operation $(2nd^{-1}, -1)$. Below the sets are the Roman numeral analyses $A: iv$ and $A: I$.

Figure 6: Multiple hearings of the opening motto of Brahms, *String Quintet in G major*, op. 111, ii, measure 1 (Rings 2011, 205)

Schenkerian analysis, and criticism (Kerman, Burnham, Žižek, Dollar) and blending them with transformational concepts. Rings brings these diverse threads with his own notation, yielding compelling observations about the interaction of harmony, hypermeter, and text setting.

[18] In fact, Rings rarely restricts himself to a single set of analytical tools. Figure 6, taken from the beginning of the Brahms Quintet analysis in Chapter Seven, is representative of his methodological pluralism. The diagram demonstrates how the opening motto can be heard in two different keys, and explores the implications of the two hearings with *Stufentheorie* (a b), Schenkerian notation (c d), Riemannian functions (e-f, expressed in Rings' oriented diagram notation), and tonal GIS sets (g h). Throughout the book, analyses that don't employ at least two different theoretical approaches are the exception rather than the rule. If, as Henry Klumpenhouwer argues, the implicit formal argument

embedded in the narrative structure of Lewin's *GMIT* aims to move the reader from an intervallic (GIS) to a transformational perspective, then the implicit argument of *Tonality and Transformation* is that one should choose one's theoretical tools wisely. In fact, "implicit" may well be an inappropriate word with which to describe this aspect of the book, for Rings states his positions quite clearly and on numerous occasions, in descriptions of his preferred "prismatic" methodology, and his injunction that because "tonal intentions at any instant are multiple and perhaps partly inchoate ... no single analytical representation can do full justice to our pre-reflective intentional activity."²² The key to this stance seems to be the frequent characterization of theoretical approaches as "technologies," implying that they are to be thought of not as ideological stances or *Weltanschauungen* (though we must be aware of the baggage they bear), but rather as interchangeable tools with particular strengths and weaknesses, to be used for sharpening and focusing our musical apperceptions.²³

[19] The analytical case studies enact this stance vividly; while they all center on the transformational methods laid out in the opening chapters, Rings never hesitates to mix in other tools where necessary. His pluralism implies that in the second decade of the twenty-first century, the discipline can no longer abide facile mappings of certain theoretical approaches onto specific repertoires: Schenker for 18th- and early 19th-century music, Neo-Riemannian theory for the late 19th century, pitch-set theory for the 20th century, and so forth. Theorists should instead choose the analytical technology that allows them to express their own musical apperceptions most clearly, without regard for the artificial divisions between subdisciplines.

[20] The comprehensive, flexible nature of Rings' ideas is both their greatest strength, and *Tonality and Transformation's* most significant liability as a self-contained study. By the end of the third chapter, well over halfway through the book, things still feel preliminary. Rings has built a huge system, re-examining and reframing much of tonal theory essentially from square one. But as a result, the book feels somewhat out of proportion: after 150 pages of theory, 70 pages of the application of that theory can hardly scratch the surface of its possibilities. Although Rings cautions that tonal-transformation theory is best suited to brief passages rather than entire movements, it might still be instructive to see, for example, how it would tackle larger forms, if not an entire sonata movement. And concerns of scope aside, the core of the symphonic repertoire is conspicuous in its absence: Haydn is never mentioned, and Beethoven appears only in tiny fragments. The book in fact seems to skirt around the repertoire that is most firmly associated with Schenkerian theory and *Formenlehre*. This could very well be intentional, as Rings' methodologies have more to gain by addressing repertoires that often fall at the fringes of traditional tonal analysis (such as Bach's E-major Fugue and the two Brahms movements). And it is interesting to note the amount of attention given to Wagner, whose music already stands as a favored subject of Neo-Riemannian theory, which prefers to investigate his music without reference to its tonality.

[21] But while the oversight of canonic tonal repertoire is understandable and forgivable, another shortcoming is perhaps less so: the lack of more than the merest whisper of hermeneutic interpretation. Rings' 2008 article "*Mystères limpides: Time and Transformation in Debussy's Des pas sur la neige*," is an

²² Rings, *Tonality and Transformation*, 106.

ambitious demonstration of the hermeneutic potential of transformational theory.²³ Rings maps an alternating ostinato onto the titular footsteps (*pas*) in the snow (*neige*), and uses this interpretive framework to tease out the barest threads of narrative in the piece and weave them into a compelling tale. Unfortunately, neither that analysis nor anything resembling its narrative insight (with the possible exception of the closing passages of Chapter Five) finds its way into the pages of *Tonality and Transformation*. While the book offers valuable insights to theorists of all stripes, and its introductory chapters will surely do a stellar job of reaching out and making mathematical and transformational ideas more accessible to graduate and undergraduate theory students, some examples of hermeneutic interpretation might have made it more attractive to musicologists as well.

[22] In the end, however, the incomplete explorations of the analytical sections do little to undermine the impressive accomplishments of the theoretical ones, and in fact seem to perform a great disciplinary service by leaving the door wide open for extensions and applications from other scholars. The book provides both a varied collection of theoretical tools, and several excellent examples for their use. There is no doubt that the coming years will see a deluge of *Tonality and Transformation*-related papers at conferences and in journals, as music theorists digest, apply, and extend the ideas presented in the book. In fact, such continued discussion—the filling-in of blanks and the exploration of the roads not taken—seems to be the highest compliment that an academic discipline can pay to a text, and it is one that *Tonality and Transformation* richly deserves.

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²³ Steven Rings, "Mystères limpides: Time and Transformation in Debussy's *Des pas sur la neige*," *19th-Century Music* 32, no. 2 (Fall 2008): 178–208.

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